Problem Set VII, Macroeconomics III
University of Copenhagen

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Exercise 1

Assume that the labor market is described by a Diamond-Mortensen-Pissarides model as discussed in the lecture notes. Unemployed workers u meet vacancies v posted by firms according to a matching function $M(u, v) = u^{\gamma} v^{1-\gamma}$.

(a) Provide an expression for the probability that a worker will find a firm $f(\theta)$ and one for the probability that a firm will meet a worker $\eta(\theta)$. The variable $\theta = v/u$ describes the tightness of the labor market.

(b) Each period, a share δ of all matches terminate for exogenous reasons. At the same time, $f(\theta)$ of all unemployed workers find a firm to work with. Derive an expression for the steady state stock of unemployed workers, u, as a function of θ. Draw the function in $u - v$ space. How are the two related? Explain.

A match between a firm and a worker produces output y. The worker receives wage w from the firm. Firms and workers discount the figure at $\frac{1}{1+r}$. A firm that is not currently matched to a worker can post a vacancy at cost κ. The discounted value of a firm that is matched with a worker is

$$J(w) = y - w + \frac{1}{1+r}[(1-\delta)J(w) + \delta V]$$

The value of a firm that has just posted a vacancy is

$$V = -\kappa + \frac{1}{1+r}\left[\eta(\theta)J(w) + (1-\eta(\theta))V\right]$$

(c) Assume entry for firms is free, any firm can choose to post a vacancy. This implies the vacancy creation curve $w = y - \frac{r+\delta}{\eta(\theta)}\kappa$. Show that this is true and explain why wages fall if θ rises.

Workers in the model are either employed or unemployed. The unemployed receive unemployment benefits b. Utility is linear.

(d) Write down a recursive expression for the value of a worker who is matched to a firm, $W(w)$, and the value of a worker who is not, U. Explain the economic intuition behind both equations.

When a firm meets a worker, the two bargain over the wage according to Nash bargaining, as discussed in the lecture: they split the total surplus $\Omega = J(w) + W(w) - U$ according to their respective bargaining weights. The worker's bargaining weight is ξ. It can be shown that this leads to the following wage curve:

$$w = \xi\kappa\theta + \xi y + (1 - \xi)b$$

(e) Explain the economic meaning behind the wage curve. Why does κ enter into the wage bargaining?

Equilibrium in the labor market is governed by the vacancy-posting curve and the wage curve.

(f) Compute the equilibrium in the labor market as a function of the model's parameters. Draw a diagram of the two curves, with θ on the x-axis and w on the y-axis. Mark the equilibrium. What happens if the separation probability, δ increases?

(g) In a search model such as this, the matching process introduces two inefficiencies. Explain each in turn.

The social planner's optimality condition is (no need to derive it):

$$(y - b)(1 - \gamma) = \kappa\frac{\delta + r}{\eta(\theta)} + \kappa\theta\gamma$$

(h) What does this tell you about the optimal value of worker bargaining power ξ? How does it relate to the two externalities discussed above?